

User Interface

Unit-6

User Interface

- User Interface should be designed to match skills, experience and expectation of its users.
- System users often judge a system by its interface rather than its functionality
- A poorly designed interface can cause a user to make catastrophic errors.
- Poor user interface design is the reason why so many software system are never used.

Multimedia User Interface

- Computer interface that communicate with users using multiple media
- Media determine how and how well human-computer interaction occurs (decisive for user acceptance)

Human Factors in Interface Design

- Limited Short-term Memory
 - People can instantaneously remember about 7 items of information. If you present more than this, they are more liable to make mistake.
- People make mistakes
 - When people make mistakes and systems go wrong, inappropriate alarms and message can increase stress and hence the likelihood of more mistake.
- People are different
 - People have a wide range of physical capabilities.
- People have different interaction preference
 - Some like pictory, some like text.

UI Design Principles

- User familiarity
 - The interface should be based user-oriented terms and concepts rather than computer concepts.
 - For example, an office system should use concepts such as letters, documents, folders etc. rather than directories, file identifiers, etc.
- Consistency
 - The system should display an appropriate level of consistency.
 - Commands and menus should have the same format etc.

UI Design Principles

- Recoverability
 - The system should provide some resilience to user errors and allow the user to recover from errors.
 - This might include an undo facility, confirmation of destructive actions, 'soft' deletes, etc.
- Minimal Surprise
 - If a command operates in a known way, the user should be able to predict the operation of comparable commands

UI Design Principles

- User Guidance
 - Some user guidance such as help systems, online manuals, etc. should be supplied.
- User Diversity
 - Interaction facilities for different types of user should be supported.
 - For example, some users have seeing difficulties and so larger text should be available

Basic Design Issues

- Two problems must be addressed in interactive system design.
 - How should information from the user be provided to computer system
 - How should information from the computer be presented to the user.
- User interaction and information presentation may be integrated through a coherent framework such as UI metaphor

Basic Design Issues

- Architectural issues
- Information Characteristics for presentation
- Presentation Function
- Presentation Design Knowledge
- Effective human Computer interface

Architectural Issues

- An effective presentation design should be as interactive as it is informative.
- The user should have the freedom to choose the direction of navigation.
- This should be supported by user-oriented goals, context sensitive help and selection of proper media in order to represent the information

Information Characteristics for presentation

- The complete set of information characteristics makes knowledge definition and representation easier because it allows for appropriate mapping between information and presentation technology.
- The information characteristics specify:
 - **Types:** Information types refer to the different forms and categories of data that need to be presented.

Numerical data (integers, decimals, percentages)

Categorical data (groups, classifications)

Temporal data (dates, times, sequences)

Binary data (yes/no, true/false)

Multimedia data (images, audio, video)

Information Characteristics for presentation

- **Relation Structure:** This describes how different pieces of information are connected and organized:

Hierarchical relationships (parent-child, tree structures)

Network relationships (many-to-many connections)

Sequential relationships (ordered data)

Dependency relationships (when one element relies on another)

- **Multi-domain Relation:** Relations can be considered across multiple domains, such
 1. Multiple attributes of a single object set (e.g.positions, colours, shapes, and/or sizes of a set of objects in a chart);
 2. Multiple object sets (e.g., a cluster of text and graphical symbols on a map)
 3. Multiple displays

Information Characteristics for presentation

- **Large Datasets:** Large datasets refers to numerous attributes of collections of heterogeneous objects. (for e.g: databases with numerous object types and attributes of technical documents for large systems etc.)

Presentation Function

- Presentation function is a program which displays an object (e.g. printf for a display of a character).
- It is important to specify the presentation function independent from presentation form, style or the information it conveys.
- One approach of it is the set of information-seeking goals and other is hierarchical representation of media-independent presentation goals derived from a plan-based theory of communication.

Presentation Design Knowledge

This encompasses the principles and best practices for creating effective interfaces, including:

- Content Selection
- Media Selection
- Color theory and usage
- Typography and readability
- Layout principles
- Visual balance and harmony
- Cultural considerations
- Industry standards and conventions

Effective Human-computer interaction

- One of the most important issues regarding multimedia interfaces is effective human-computer interaction of the interface, i.e., user-friendliness.
- Here are the main issues the user interface designer should keep in mind:
 1. Context: Context refers to how well the interface fits into the user's environment and needs.
 2. Linkage to the world beyond the presentation display;
 3. Evaluation of the interface with respect to other human-computer interfaces;
 4. Interactive capabilities, and
 5. Separability of the user interface from the application.

Video at User Interface

- Video is actually the continuous sequence of still images such that the rate of replacement of images is 15 images per second.
- When incorporating video into a UI, designers must consider:
 - **Performance:** Video files can be large and resource-intensive. Optimize for fast loading and smooth playback.
 - **Responsiveness:** Ensure videos adapt to different screen sizes and resolutions.
 - **Accessibility:** Provide captions, transcripts, and audio descriptions for users with disabilities.
 - **User Control:** Include play/pause, volume, and full-screen controls for a better user experience.

Video at User Interface

Hardware for Visualization of Motion Pictures

- Special hardware for visualization of motion pictures is available today, mostly through additional video cards.
- Early examples of such additional hardware are IBM-M-Motion and ActionMedia II (Intel/IBM) cards, and the Parallax, Sun and RasterOps cards.
- These cards were designed to handle the demanding tasks of video processing, including decoding, rendering, and displaying motion pictures
- Today, the functionality of these specialized cards has largely been integrated into mainstream graphics processing units (GPUs) and multimedia systems.

Video at User Interface

Example of Video in UI

- E-Learning Platforms
- Streaming Services
- Marketing Websites
- Gaming Interfaces

Audio at User Interface

- Audio can be implemented at the user interface for application control.
- Thus, speech analysis is necessary. Speech analysis is either speaker-dependent or speaker-independent.
- **Speaker-dependent analysis** are trained to recognize the speech patterns of a specific user. They require the user to "train" the system by providing samples of their voice.
- **Speaker-independent analysis** are designed to recognize speech from any user without requiring prior training. They rely on generalized models of speech patterns. Basic voice assistants (e.g., Siri, Google Assistant).

Audio at User Interface

- During audio output, the additional presentation dimension of space can be introduced using two or more separate channels to give a more natural distribution of sound.
- The best known example of this is stereo.

Audio at User Interface

- The concept of the audio window allows for application independent control of audio parameters, including spatial positioning.
- Most current multimedia applications using audio determine the spatial positioning themselves and do not allow the user to change it.
- An example of such an application is the audio tool for SUN workstations.

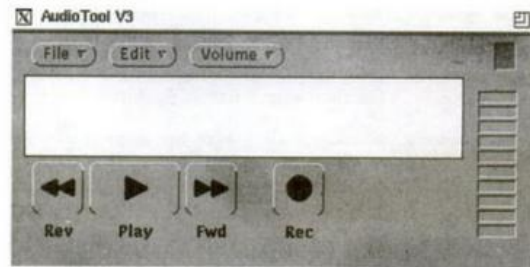


Figure 14.5: Audio tool user interface.

User Friendliness as the Primary Goal

- User friendliness is the main property of a good user interface.
- The design of user-friendly graphical interface requires the consideration of many conditions.
- The addition of audio and video to the user interface does not simplify this process.
- The user-friendliness is implemented as
 - Easy to learn instructions
 - Presentation
 - Dialogue Boxes
 - Additional Design Criteria
 - Design Specific Criteria

Easy to learn instruction

- The instructions guiding the use of interface should be easy to learn. The language should be simple and graphical.
- The older dial phones required no time to learn. An ISDN telephone requires more time as compared to simple touch phone.

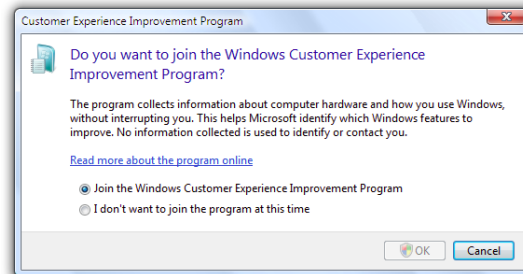
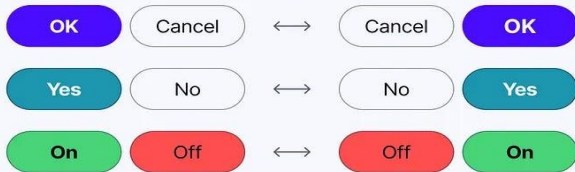


Presentation

- The presentation, i.e., the optical image at the user interface, can have the following variants:
 - Full text
 - Abbreviated text
 - Icons i.e. graphics
 - Micon i.e Motion video

Dialogue Boxes

- Different dialogue boxes should have a similar construction. This requirement applies to the design of :
 - the buttons OK and Abort
 - Joined Windows
 - Other applications in the same window system
- Semantically similar entry functions can be located in one dialogue box instead of several dialogue boxes.



Additional Design Criteria

1. Dynamic Cursor Feedback:
 - Change the cursor shape (e.g., rotating fish) to indicate system states like "task in progress."
2. Progress Visualization:
 - Show progress for time-intensive tasks (e.g., filling bar for disk formatting or bytes transferred for file retrieval).
 - Always include an Abort option to let users cancel the task.
3. Immediate Selection Feedback:
 - Highlight selected entries as "work in progress" to prevent further input until the task starts.

Design Specific Criteria

In addition to the above-described general criteria for the design of a user interface, the problem specific properties of the actual task need to be considered. These properties are demonstrated in our telephone service example. The telephone network and telephone-specific end-devices are provided by the telephone companies. They specify the user interface characteristics

- In multimedia systems, emulate a key-based interface for compatibility.
- Use a telephone icon to indicate active applications without occupying the entire screen.
- When a call request arrives, it must be immediately signalled (e-g., ringing).

Assignment

1. User Interface design process of multimedia with block diagram
2. Application Development Life cycle of multimedia system

Application of the Multimedia Development Life Cycle (MDLC)

Methodology to Build a Multimedia-Based Learning System